Unidade - 09

Cálculo de Integrais.

O Problema

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$$F'(x) = f(x)$$

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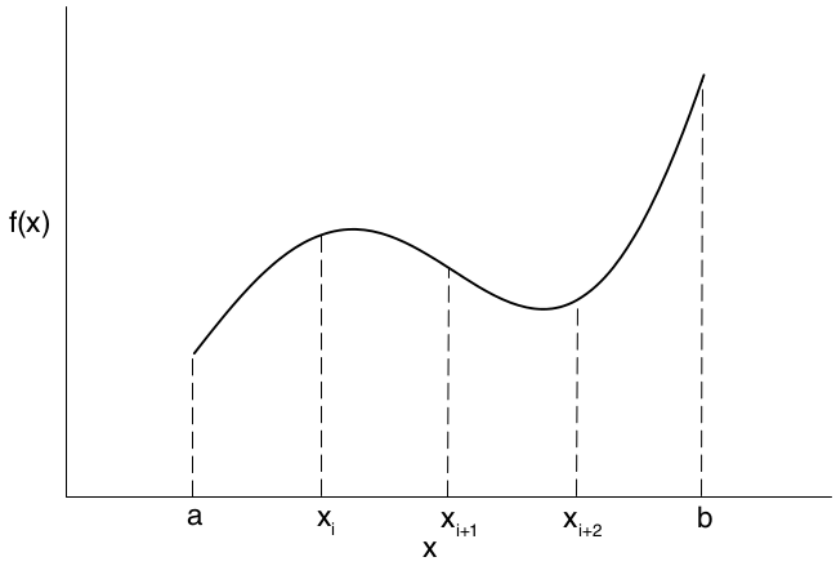

$$x_1 = a$$

$$x_N = b$$

$$x_i = a + (i-1)h$$

$$h = \frac{b-a}{N-1}$$

Em geral

$$I = \int_a^b f(x)dx \approx \sum_{i=1}^{N} f(x_i)w_i$$

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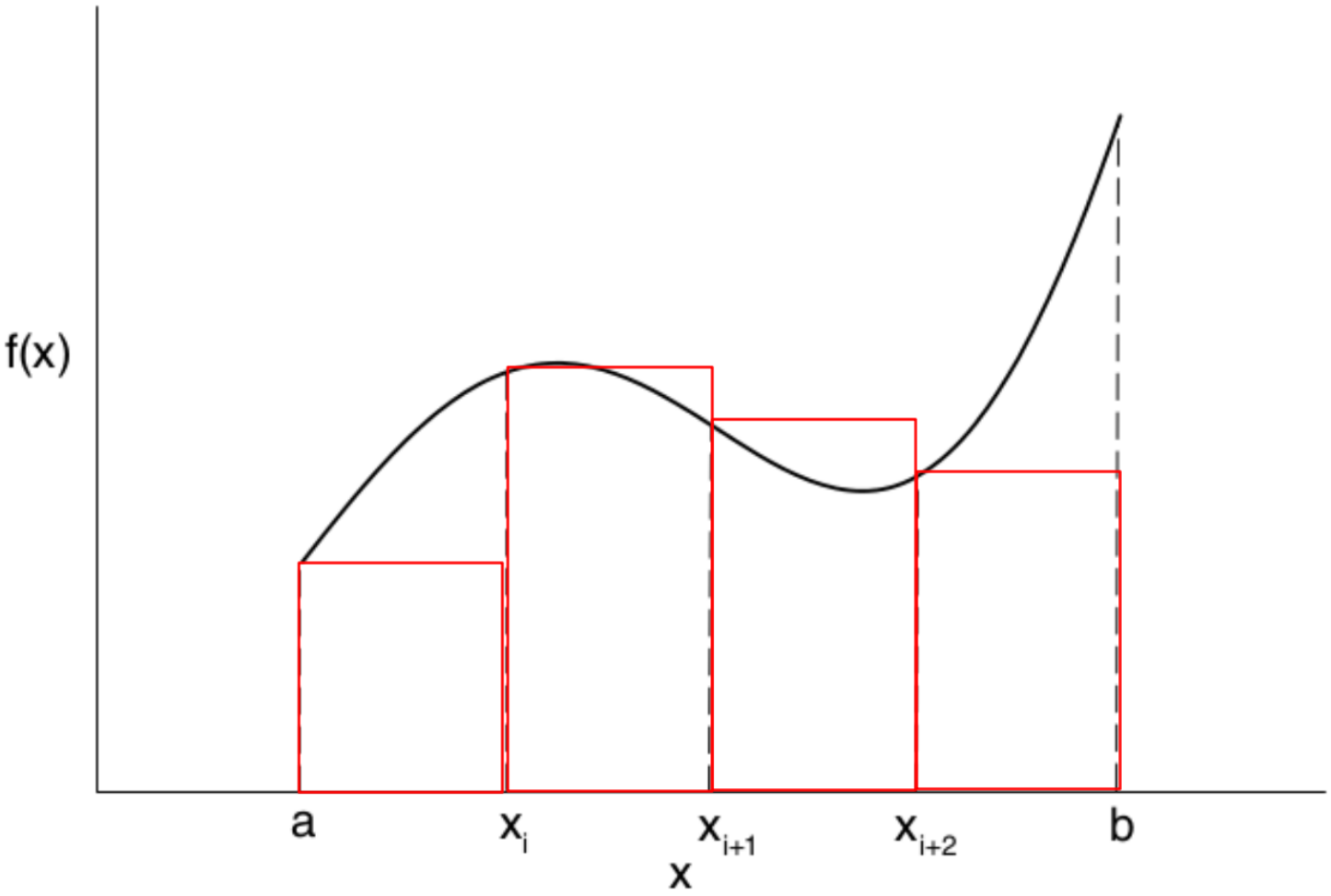
Onde os w_i dependem do método escolhido.

Valor à esquerda do intervalo



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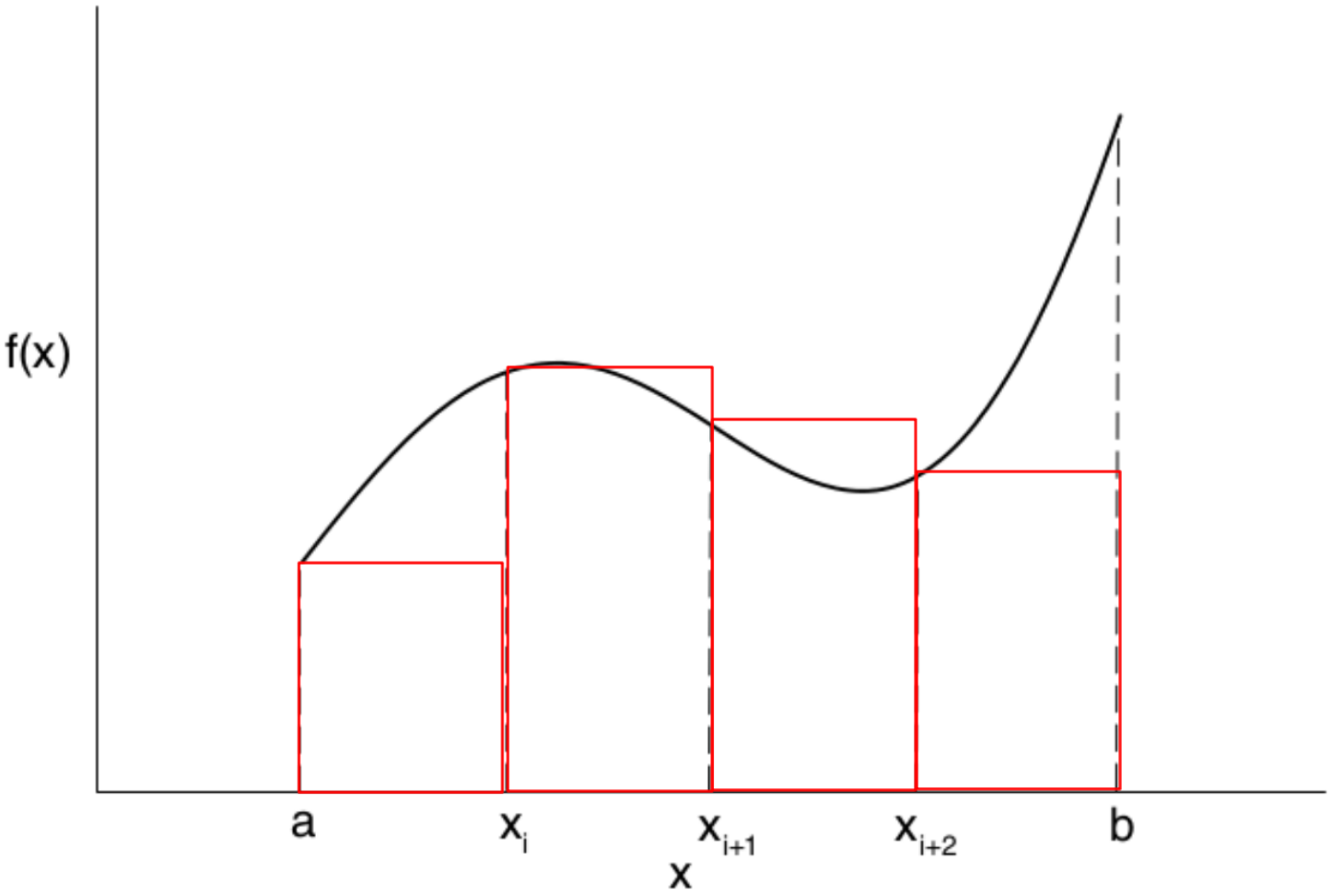
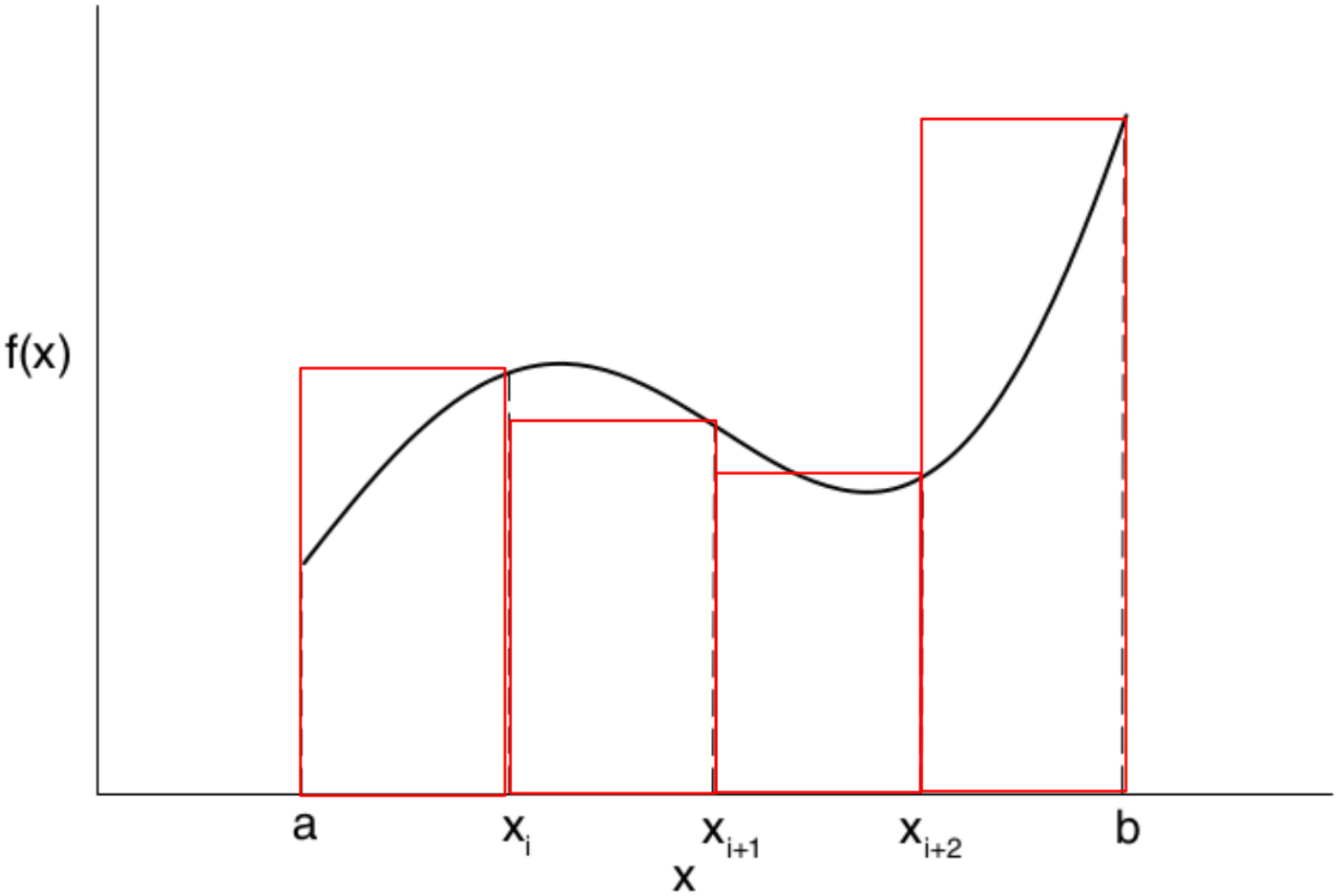
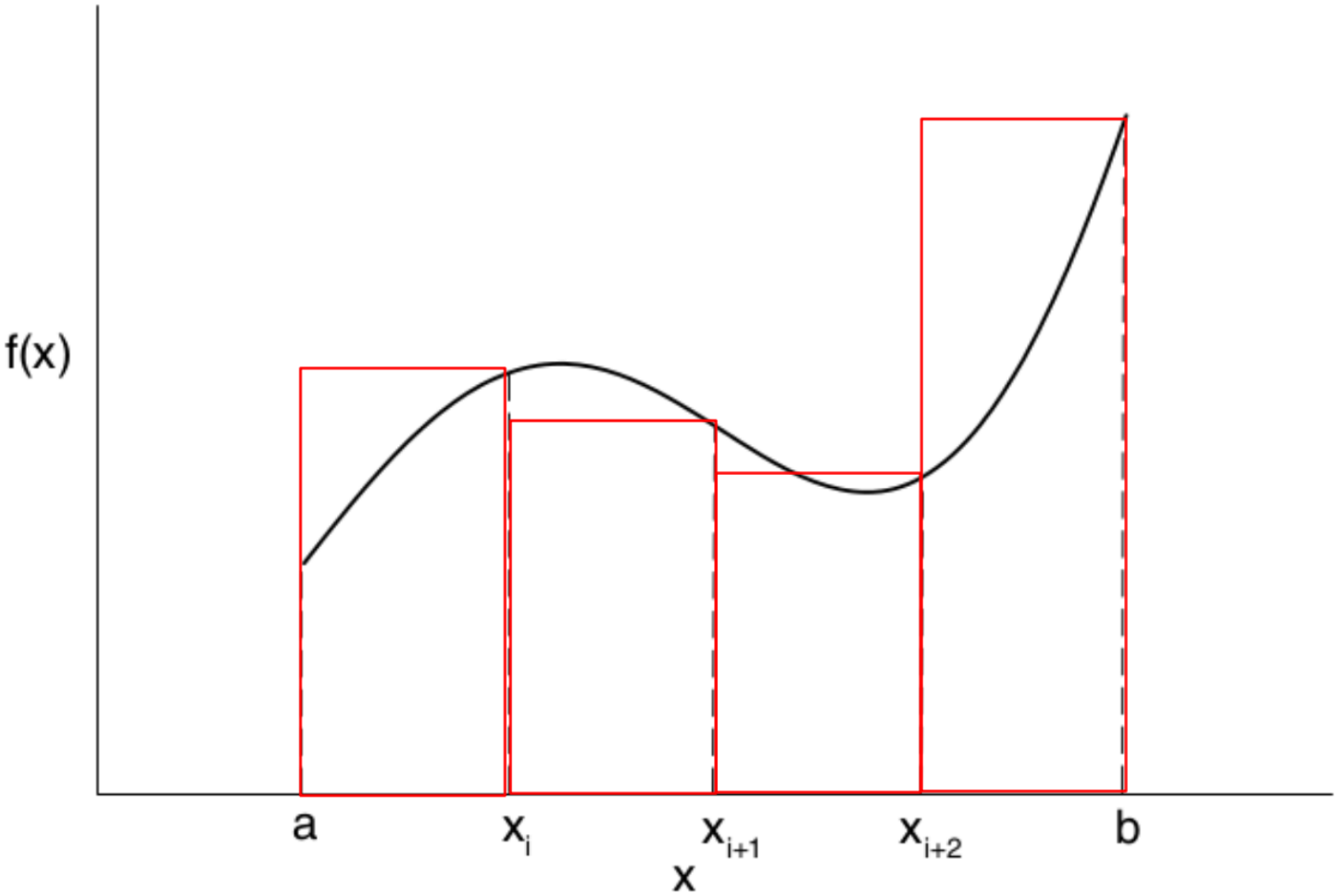


$$I = \sum_{i=1}^{N} f(x_i) w_i \qquad \begin{array}{rcll} w_i & = & h & \rightarrow \quad i = 1, 2, 3, ..., N-1 \\ w_i & = & 0 & \rightarrow \quad i = N \end{array}$$

Valor à direita do intervalo



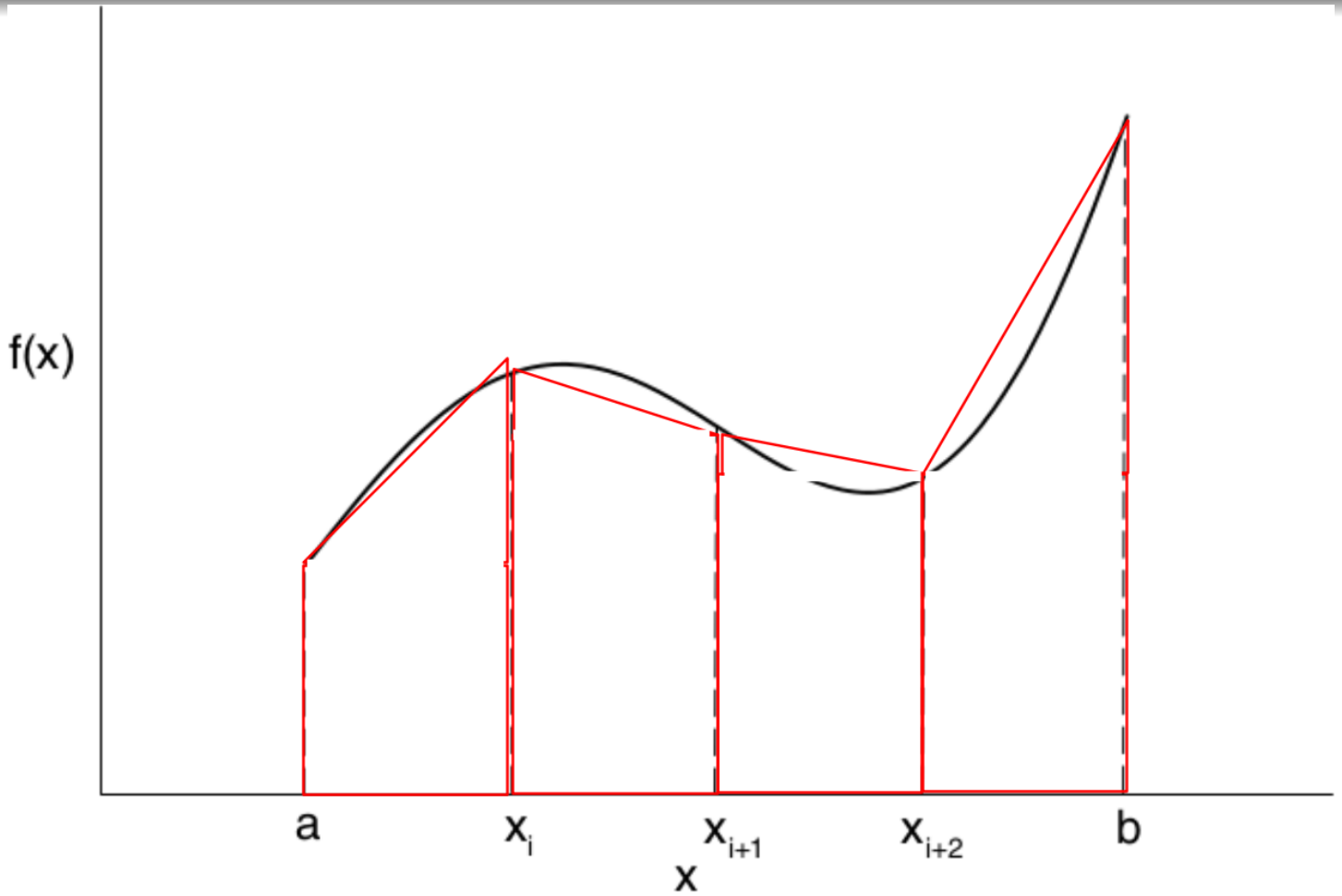
Valor à direita do intervalo



$$I = \sum_{i=1}^{N} f(x_i) w_i$$

$$\begin{aligned} w_i &= 0 \quad \rightarrow \quad i = 1 \\ w_i &= h \quad \rightarrow \quad i = 2, 3, ..., N \end{aligned}$$

Regra do trapézio



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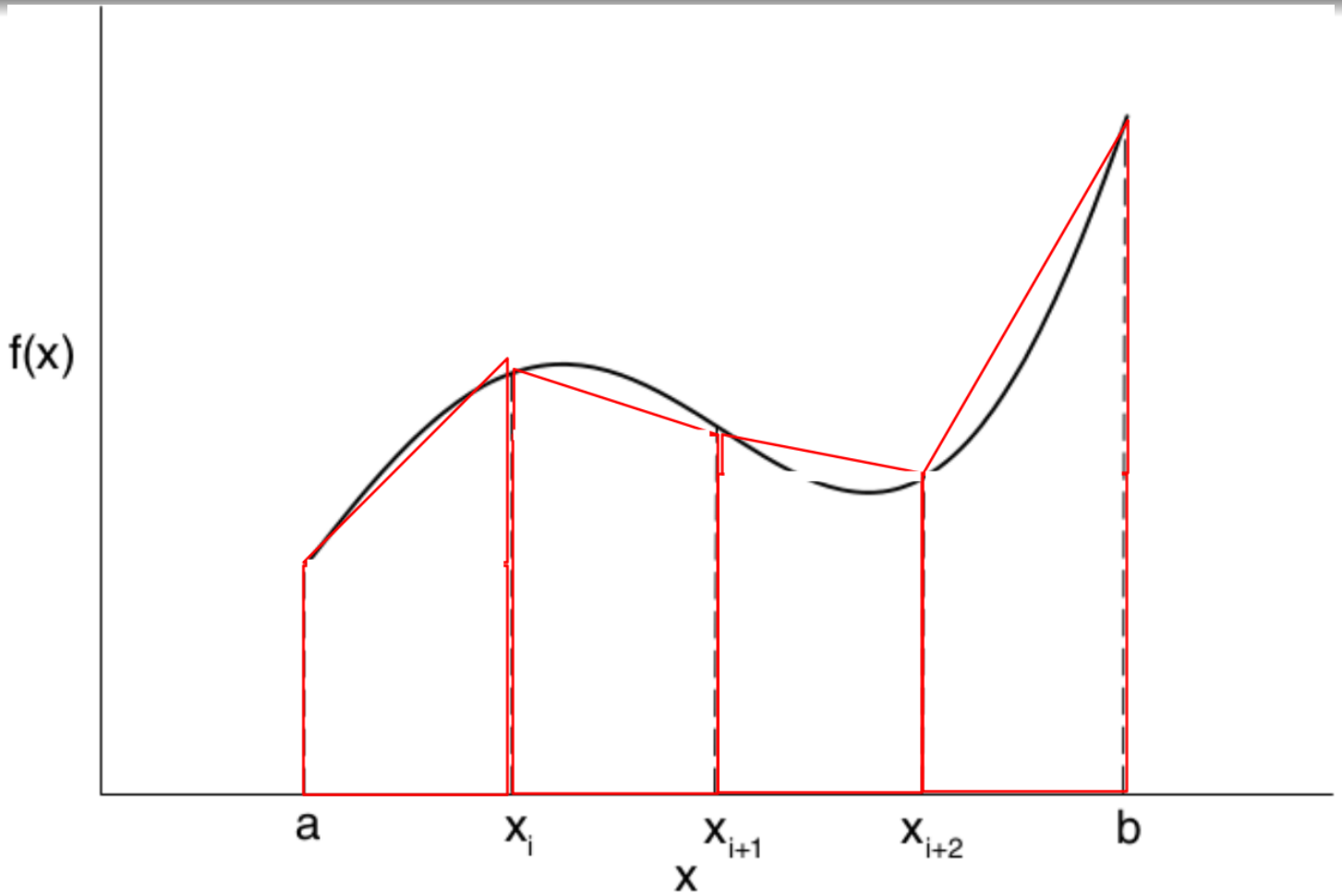


$$I = \sum_{i=1}^{N} f(x_i) w_i \qquad \begin{aligned} w_i &= h \quad \rightarrow \quad i = 2, 3, ..., N-1 \\ w_i &= \frac{h}{2} \quad \rightarrow \quad i = 1, N \end{aligned}$$

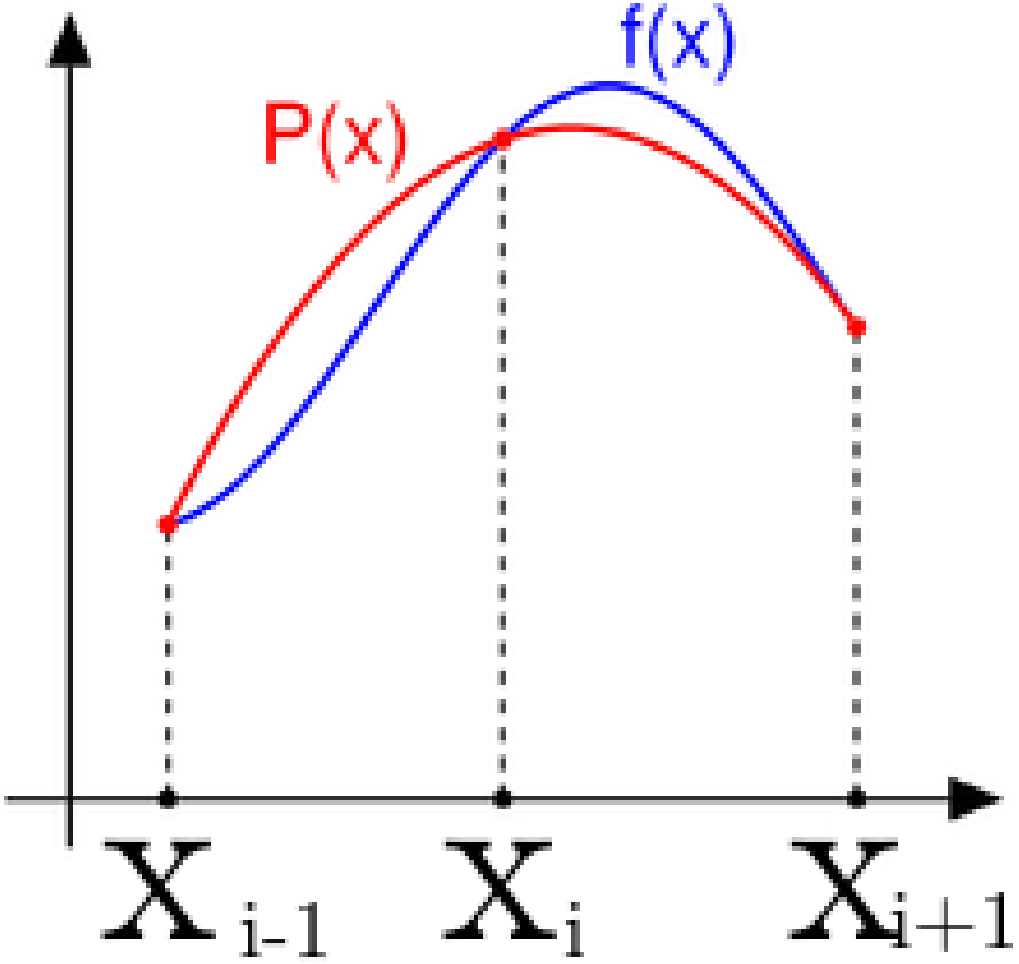
Exemplos

- Escreva programas para o cálculo das funções:

$$
\begin{aligned}
f(x) &= x^n, \quad n = 1, 2, 3, 4, \ldots \\
f(x) &= e^{\pm x}
\end{aligned}
$$

No intervalo de $x = 0$ a $x = 1$.

Regra de Simpson



Regra de Simpson

$$P(x) = \alpha x^2 + \beta x + \gamma$$

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$$\begin{aligned} P(x_i) &= \alpha x_i^2 + \beta x_i + \gamma = f(x_i) = f_i \\ x_{i-1} &= x_i - h \end{aligned}$$

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$$\int_{x_{i-1}}^{x_{i+1}} P(x)dx = \frac{1}{3}\alpha x^3\Big|_{x_{i-1}}^{x_{i+1}} + \frac{1}{2}\beta x^2\Big|_{x_{i-1}}^{x_{i+1}} + 2\gamma h$$

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$$\int_{x_{i-1}}^{x_{i+1}} P(x)dx = \left(2\alpha x_i^2 + \frac{2}{3}\alpha h^2 + 2\beta x_i + 2\gamma\right)h$$

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Isso corresponde à integração de dois subintervalos

$$x_{i-1} \longrightarrow x_i \qquad \text{e} \qquad x_i \longrightarrow x_{i+1}$$

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$$\begin{aligned} w_i &= \frac{h}{3} \quad \rightarrow \quad i = 1, N \\ w_i &= \frac{4h}{3} \quad \rightarrow \quad i = \text{par} \\ w_i &= \frac{2h}{3} \quad \rightarrow \quad i = \text{ímpar} \neq 1, N \end{aligned}$$

Polinômios Superiores

$$\int_{x_{i-1}}^{x_{i+2}} P(x)dx = \left(f_{i-1} + 3f_i + 3f_{i+1} + f_{i+2}\right)\frac{3h}{8}$$

$$\int_{x_{i-2}}^{x_{i+2}} P(x)dx = \left(7f_{i-2} + 32f_{i-1} + 12f_i + 32f_{i+1} + 7f_{i+2}\right)\frac{2h}{45}$$